

Topic: Sequence and Recursion

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Part 1 Sequence

Section 1 Introduction

- Introduction
- Definition and Notation
- Finite and Infinite Sequence

Introduction

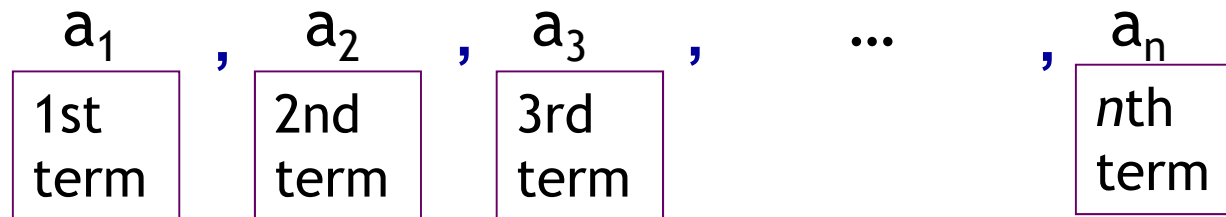
- Sometimes, there is a need to discover and characterize regular patterns e.g. repeated processes
- The mathematical *structure* used to study repeated processes → *sequence*

Examples of Sequences

- (i) 1, 3, 5, 7, . . .
[An infinite sequence of odd numbers]
- (ii) 1, 4, 9, 16, . . . , n^2
[A finite sequence of n terms
of squares of integers]
- (iii) 0, 1, 1, 2, 3, 5, 8, 13,
[An infinite sequence called Fibonacci Sequence]

Definition & Notation

A sequence is
a set of terms
written in a definite order obeying a certain rule.



Finite and Infinite Sequence

In general,

A *finite sequence* is notated as

a_m , a_{m+1} , a_{m+2} , ... , a_n

Each individual term a_k (read “a sub k ”) is called a *term*.

The k in a_k is called a subscript.

m is the subscript of the initial term; and

n is the subscript of the final term.

An *infinite sequence* is notated as

a_m , a_{m+1} , a_{m+2} , ...

Part 1 Sequence

Section 2 Explicit Sequence

- Explicit Formula
- Finding Terms of Explicit Sequence
- Alternating Sequence
- Finding Explicit Formula Given The Terms

Explicit Formula

An explicit formula for a sequence is
a general formula that shows how each a_k is related to k .
e.g. $a_k = 5k - 1$

Example (Finding Terms of Explicit Sequence)

Find the first three terms of the sequence $a_n = 5n - 1$.

Solution:

$$a_1 = 5(1) - 1 = 4$$

$$a_2 = 5(2) - 1 = 9$$

$$a_3 = 5(3) - 1 = 14$$

The first 3 terms of the sequence are 4, 9 and 14.

Exercise (Finding Terms of Explicit Sequence)

Find the first four terms of the sequence

$$a_n = 2n^2 + 3 \text{ where } n \geq 5.$$

Solution:

$$a_5 = 2(5)^2 + 3 = 53$$

$$a_6 = 2(6)^2 + 3 = 75$$

$$a_7 = 2(7)^2 + 3 = 101$$

$$a_8 = 2(8)^2 + 3 = 131$$

The first 4 terms of the sequence are 53, 75, 101, 131.

Note :

The first term starts with subscript $n=5$

Example (Alternating Sequence)

Find the first six terms of the sequence $b_n = (-1)^n$ for $n \geq 2$

Solution:

$$b_2 = (-1)^2 = 1$$

$$b_5 = (-1)^5 = -1$$

$$b_3 = (-1)^3 = -1$$

$$b_6 = (-1)^6 = 1$$

$$b_4 = (-1)^4 = 1$$

$$b_7 = (-1)^7 = -1$$

The first 6 terms of b_n are $1, -1, 1, -1, 1, -1$

Note :

Even powers yield positive values. Odd powers yield negative values.

The sequence oscillates endlessly between positive and negative values.

Exercise (Alternating Sequence)

Find the first four terms of the sequence $c_n = \left\{ \frac{(-1)^n}{1+2n} \right\}$ for $n \geq 0$

Solution:

$$c_0 = \left\{ \frac{(-1)^0}{1+2(0)} \right\} = 1$$

$$c_2 = \left\{ \frac{(-1)^2}{1+2(2)} \right\} = \frac{1}{5}$$

$$c_1 = \left\{ \frac{(-1)^1}{1+2(1)} \right\} = -\frac{1}{3}$$

$$c_3 = \left\{ \frac{(-1)^3}{1+2(3)} \right\} = -\frac{1}{7}$$

The first 4 terms of c_n are

$$1, -\frac{1}{3}, \frac{1}{5}, -\frac{1}{7}$$

Note :

Even powers yield positive values. Odd powers yield negative values.

The sequence oscillates endlessly between positive and negative values.

Example (Finding Explicit Formula Given The Terms)

Find an explicit formula for a sequence that has the following initial terms:

1, 4, 9, 16, 25, 36, ...

Solution:

Step 1 : Denote each term by a_k ,
assuming 1st term is a_1 .

1,	4,	9,	16,	25,	36
↕	↕	↕	↕	↕	↕
a_1	a_2	a_3	a_4	a_5	a_6

Step 2 : Write each term in term of k :
(where k is the number in the subscript)

↕	↕	↕	↕	↕	↕
1^2 ,	2^2 ,	3^2 ,	4^2 ,	5^2 ,	6^2

Step 3 : Write down general formula.

$a_k = k^2 \quad \text{for } k \geq 1$
--

Example (Finding Explicit Formula Given The Terms)

Make use what you learnt in Examples 2.1.2/2 and 2.1.2/3 to find an explicit formula for a sequence that has the following initial terms:

-1, 4, -9, 16, -25, ...

Solution:

Step 1 : Denote each term by a_k ,
assuming 1st term is a_1 .

-1,	4,	-9,	16,	-25
$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$
a_1	a_2	a_3	a_4	a_5

Step 2 : Write each term in terms of k .

$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$
$-1^2,$	$2^2,$	$-3^2,$	$4^2,$	-5^2

Step 2b : Insert the factor $(-1)^k$ to get ± 1 .

$(-1)^1(1^2)$	$(-1)^2(2^2)$	$(-1)^3(3^2)$	$(-1)^4(4^2)$	$(-1)^5(5^2)$
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Step 3 : Write down general formula.

$a_k = (-1)^k k^2 \quad \text{for } k \geq 1$

Part 2 Summation Notation

Section 1 Introduction

- Introducing Summation Notation
- Definition and Notation

Introducing Summation Notation

Question

What if you need to add up all the terms in a sequence?

Σ = Greek letter sigma = sum or summation

For example,

$$\sum_{k=1}^n a_k = a_1 + a_2 + a_3 + \dots + a_n$$

Hence, $\sum_{k=1}^n a_k$ is a shorthand for the sum of a_1 to a_n .

Definition & Notation

In general, the sum of a sequence of terms $a_m, a_{m+1}, a_{m+2}, \dots, a_n$ (where m and n are integers and $m \leq n$) is denoted by :

$$\sum_{k=m}^n a_k = a_m + a_{m+1} + a_{m+2} + \dots + a_n$$

k = an index of the summation = counter

m = lower limit of the summation

n = upper limit of the summation

Sometimes when solving problems, it might be helpful to manipulate the summation notation by separating off the final term or adding a final term to the summation as follows:

$$\sum_{k=m}^n a_k = \sum_{k=m}^{n-1} a_k + a_n$$

Part 2 Summation Notation

Section 2 Computing Summations

- Finding Summation Given Individual Terms
- When The Terms are Given By Formula
- Changing From Summation Notation to Expanded Form
- Changing From Expanded Form to Summation Notation

Example

(Finding Summations Given Individual Terms)

Given $t_1 = 3$, $t_2 = 6$, $t_3 = 4$, $t_4 = -3$, $t_5 = -1$, compute :

$$\text{a. } \sum_{k=2}^4 t_k = t_2 + t_3 + t_4 = 6 + 4 + (-3) = 7$$

$$\text{b. } \sum_{k=3}^3 t_k = t_3 = 4$$

$$\text{c. } \sum_{k=1}^2 t_{2k} = t_{2 \times 1} + t_{2 \times 2} = t_2 + t_4 = 6 + (-3) = 3$$

Example (When the Terms Are Given by a Formula)

$$\sum_{k=4}^7 (2k - 3)$$

$$= [2(4) - 3] + [2(5) - 3] + [2(6) - 3] + [2(7) - 3]$$

$$= 5 + 7 + 9 + 11 = 32$$

$$\sum_{k=4}^7 2k - 3$$

Does the Σ sign include “-3”?

$$= [2(4) + 2(5) + 2(6) + 2(7)] - 3$$

$$= [8 + 10 + 12 + 14] - 3 = 41$$

Example

(Changing from Summation Notation to Expanded Form)

Evaluate

$$\sum_{m=k}^{k+3} \frac{2}{3m+1}$$

$$= \frac{2}{3(k)+1} + \frac{2}{3(k+1)+1} + \frac{2}{3(k+2)+1} + \frac{2}{3(k+3)+1}$$

$$= \frac{2}{3k+1} + \frac{2}{3k+4} + \frac{2}{3k+7} + \frac{2}{3k+10}$$

Example (Changing from Expanded Form to Summation Notation)

Express $\frac{1}{n+1} + \frac{2}{n+2} + \frac{3}{n+3} + \dots + \frac{n}{2n}$ using summation notation.

Solution :

$$\begin{array}{ccccccc} \textcircled{1} & & \textcircled{2} & & \textcircled{3} & & n \\ \hline \frac{1}{n+1} & + & \frac{2}{n+2} & + & \frac{3}{n+3} & + \dots + & \frac{n}{2n} \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ k=1 & & k=2 & & k=3 & & k=n \text{ (since denominator } 2n = n+n) \end{array}$$

The explicit formula can be expressed as $\frac{k}{n+k}$ for integer k from 1 to n .

$$\text{Hence, } \frac{1}{n+1} + \frac{2}{n+2} + \frac{3}{n+3} + \dots + \frac{n}{2n} = \sum_{k=1}^n \frac{k}{n+k}$$

Exercise

(Finding Summations Given Individual Terms)

Given $b_3 = -3$, $b_4 = 8$, $b_5 = 100$, compute :

$$\text{a. } \sum_{k=3}^5 b_k = b_3 + b_4 + b_5 = -3 + 8 + 100 = 105$$

$$\text{b. } \sum_{k=4}^4 (b_k + b_{k+1}) = b_4 + b_{4+1} = b_4 + b_5 = 8 + 100 = 108$$

Exercise

(When the Terms Are Given by a Formula)

$$\begin{aligned} & \sum_{k=99}^{100} (k + 1) \\ &= [99+1] + [100+1] \\ &= 100 + 101 = 201 \end{aligned}$$

$$\begin{aligned} & \sum_{k=99}^{100} k + 1 \quad \text{Does the } \Sigma \text{ sign include “+1”?} \\ &= [99 + 100] + 1 \\ &= [199] + 1 = 200 \end{aligned}$$

Exercise

(Changing from Summation Notation to Expanded Form)

Evaluate

$$\sum_{k=p-1}^{p+1} \frac{1}{k}$$

$$= \frac{1}{p-1} + \frac{1}{p} + \frac{1}{p+1}$$

Exercise

(Changing from Expanded Form to Summation Notation)

Express $(n-1) + (n-2) + (n-3) + \dots + 1$ using summation notation.

Solution :

$$\begin{array}{ccccccc} (n-1) & + & (n-2) & + & (n-3) & + & \dots + 1 \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ k=1 & & k=2 & & k=3 & & k=n-1 \text{ (since } n-k = n - (n-1) = 1 \text{)} \end{array}$$

The explicit formula can be expressed as $n-k$ for integer k from 1 to $n-1$.

Hence, $(n-1) + (n-2) + (n-3) + \dots + 1 = \sum_{k=1}^{n-1} (n-k)$

Brackets!

Part 2 Summation Notation

Section 3 Change of Variables

- Index of Summation
- Transforming Appearance Without Changing Effect
- Transforming Summation By Change of Variables

Index of Summation

The symbol (k) used to represent the index of a summation can be changed to some other symbols without changing the effect.

For example, given these 2 summations :

$$\sum_{k=1}^4 (k+1)^2 = \underset{\substack{\uparrow \\ k=1}}{(1+1)}^2 + \underset{\substack{\uparrow \\ k=2}}{(2+1)}^2 + \underset{\substack{\uparrow \\ k=3}}{(3+1)}^2 + \underset{\substack{\uparrow \\ k=4}}{(4+1)}^2 = 54$$

$$\sum_{p=1}^4 (p+1)^2 = \underset{\substack{\uparrow \\ p=1}}{(1+1)}^2 + \underset{\substack{\uparrow \\ p=2}}{(2+1)}^2 + \underset{\substack{\uparrow \\ p=3}}{(3+1)}^2 + \underset{\substack{\uparrow \\ p=4}}{(4+1)}^2 = 54$$

Hence,
$$\sum_{k=1}^4 (k+1)^2 = \sum_{p=1}^4 (p+1)^2$$

Transforming Appearance Without Changing Effect

The appearance of the summation can also be transformed without changing the effect.

For example, given these 2 summations :

$$\sum_{k=1}^4 (k+1)^2 = (1+1)^2 + (2+1)^2 + (3+1)^2 + (4+1)^2 = 2^2 + 3^2 + 4^2 + 5^2$$

$$\sum_{p=2}^5 p^2 = 2^2 + 3^2 + 4^2 + 5^2$$

Hence, $\sum_{k=1}^4 (k+1)^2 = \sum_{p=2}^5 p^2$ although they appear to be different.

So how do I perform a change of variables?

Example

(Transform Summation by Change of Variable)

Given : $\sum_{k=1}^4 (k+1)^2$, change the index to the variable : $p = k+1$

Solution :

Step 1 : Calculate the lower and upper limits of the new summation.

When $k=1$, $p = 1+1 = 2$

When $k=4$, $p = 4+1 = 5$

Step 2 : $p = k+1 \Rightarrow k = p-1$.

Now, replace all the k in the explicit formula with $p-1$.

$$(k+1)^2 = (p-1+1)^2 = p^2$$

$$\text{Hence, } \sum_{k=1}^4 (k+1)^2 = \sum_{p=2}^5 p^2 \quad [\text{Check : } \sum_{p=2}^5 p^2 = 2^2 + 3^2 + 4^2 + 5^2 = 54]$$

Exercise (Transform Summation by Change of Variable)

Given : $\sum_{k=5}^8 k^5$, change the index to the variable : $p = k-4$

Solution :

Step 1 : Calculate the lower and upper limits of the new summation.

When $k=5$, $p = 5-4 = 1$

When $k=8$, $p = 8-4 = 4$

Step 2 : $p = k-4 \Rightarrow k = p+4$.

Now, replace all the k in the explicit formula with $p+4$.

$$k^5 = (p+4)^5$$

Hence,
$$\sum_{k=5}^8 k^5 = \sum_{p=1}^4 (p+4)^5 \quad \left[\text{Check : } \sum_{p=1}^4 (p+4)^5 = 5^5 + 6^5 + 7^5 + 8^5 \right]$$

Part 3 Product Notation

Section 1 Introduction

- Introducing Product Notation
- Definition & Notation

Introducing Product Notation

The notation of the product of a sequence has the same idea as the notation of the summation.

$\prod$ = Greek letter pi = product

For example,

$$\prod_{k=1}^n a_k = a_1 \times a_2 \times a_3 \times \dots \times a_n$$

Hence, $\prod_{k=1}^n a_k$ is a shorthand for the product of a_1 to a_n .

Definition and Notation

In general, the product of a sequence of terms $a_m, a_{m+1}, a_{m+2}, \dots, a_n$ (where m and n are integers and $m \leq n$) is denoted by :

$$\prod_{k=m}^n a_k = a_m \times a_{m+1} \times a_{m+2} \times \dots \times a_n$$

k = an **index** of the summation = counter

m = **lower limit** of the summation

n = **upper limit** of the summation

Sometimes when solving problems, it might be helpful to manipulate the product notation by separating off the final term or multiplying a final term to the product as follows:

$$\prod_{k=m}^n a_k = \left(\prod_{k=m}^{n-1} a_k \right) \bullet a_n$$

Part 3 Product Notation

Section 2 Computing Products

- Finding Products Given Individual Terms
- When The Terms are Given By Formula
- Changing From Product Notation to Expanded Form
- Changing From Expanded Form to Product Notation
- Transforming Products by Change of Variables

Computing Products

You can now apply everything you have learnt under the section for summation to the product notation!

Example (When The Terms are Given By Formula)

Compute the product of $\prod_{k=1}^4 \frac{k+1}{k}$

Solution:

$$\prod_{k=1}^4 \frac{k+1}{k} = \begin{array}{ccccccc} \begin{array}{c} \cancel{2} \\ \hline 1 \end{array} & \begin{array}{c} \cancel{3} \\ \hline \cancel{2} \end{array} & \begin{array}{c} \cancel{4} \\ \hline \cancel{3} \end{array} & \begin{array}{c} \cancel{5} \\ \hline \cancel{4} \end{array} & = & \begin{array}{c} 5 \\ \hline 1 \end{array} & = 5 \\ \uparrow & \uparrow & \uparrow & \uparrow & & & \\ k=1 & k=2 & k=3 & k=4 & & & \end{array}$$

Exercise

(Finding Products Given Individual Terms)

Given $b_3 = -3$, $b_4 = 8$, $b_5 = 100$, compute :

$$\text{a. } \prod_{k=3}^5 b_k = b_3 \cdot b_4 \cdot b_5 = (-3)(8)(100) = -2400$$

$$\text{b. } \prod_{k=3}^4 b_{k+1} = b_{3+1} \cdot b_{4+1} = b_4 \cdot b_5 = (8)(100) = 800$$

Exercise (When the Terms Are Given by a Formula)

Compute the product of $\prod_{k=4}^5 \frac{k+1}{k} - 1$

Solution:

$$\prod_{k=4}^5 \frac{k+1}{k} - 1 = \left(\frac{\cancel{5}}{4} \right) \left(\frac{6}{\cancel{5}} \right) - 1 = \left(\frac{6}{4} \right) - 1 = \frac{1}{2}$$

$\uparrow \quad \uparrow$
 $k=4 \quad k=5$

Careful! Does the $\prod$ sign include “-1”?

Exercise

(Changing from Product Notation to Expanded Form)

Evaluate

$$\prod_{k=p}^{p+2} \frac{k}{k+1}$$
$$= \left(\frac{p}{\cancel{p+1}} \right) \cdot \left(\frac{\cancel{p+1}}{\cancel{p+2}} \right) \cdot \left(\frac{\cancel{p+2}}{p+3} \right)$$
$$= \frac{p}{p+3}$$

Exercise (Changing from Expanded Form to Product Notation)

Express $(1)(2)(3)\dots(n-1)(n)$ using product notation.

Solution :

$$\begin{array}{ccccccc} (1)(2)(3)\dots(n-1)(n) & & & & & & \\ \downarrow & \downarrow & \downarrow & & \downarrow & & \\ k=1 & & k=3 & & k=n & & \\ & \downarrow & & & & & \\ & k=2 & & & & & \end{array}$$

The explicit formula can be expressed as k for integer k from 1 to n .

$$\text{Hence, } (1)(2)(3)\dots(n-1)(n) = \prod_{k=1}^n k = n! \text{ (read as “n factorial”)} \\ \text{(used as shorthand for } \prod_{k=1}^n k \text{)}$$

Exercise

(Transform Product by Change of Variable)

Given : $\prod_{k=2}^4 k^2$, change the index to the variable : $p = k+2$

Solution :

Step 1 : Calculate the lower and upper limits of the new product.

When $k=2$, $p = 2+2 = 4$

When $k=4$, $p = 4+2 = 6$

Step 2 : $p = k+2 \Rightarrow k = p-2$.

Now, replace all the k in the explicit formula with $p-2$.

$$k^2 = (p-2)^2$$

Hence, $\prod_{k=2}^4 k^2 = \prod_{p=4}^6 (p-2)^2$ [Check : $\prod_{p=4}^6 (p-2)^2 = (2^2)(3^2)(4^2)$]

Application : One-Dimensional Arrays (Using Sequences)

Where can sequences be found in computing?

→ One-dimensional arrays

Suppose that a programmer wants to :

1. read the weights of 20 students;
2. list the weights; and
3. find the total weight of all 20 students.

To avoid using separate variable names for each weight, each of those weights can be written as a term of a one-dimensional array :

`wt[1], wt[2], ... , wt[20].`

Each of these terms in the array will be stored in memory, and can be retrieved for calculations at later stages.

Application : Dummy Variable in A Loop (Using Change of Variables)

The index in a **FOR** loop in programming is a dummy variable.

In one-dimensional arrays, the dummy variable serves as a counter to track the number position of the term as we progress through the array.

For example, to list the weights of all students, the following algorithms yield the same result :

1. For $k := 1$ to 20 print wt[k] next k	2. For $p := 0$ to 19 print wt[p+1] next p	3. For $q := 2$ to 21 print wt[q-1] next q
--	--	--

Change of
variable using
 $p = k - 1$

Change of
variable using
 $q = k + 1$

Application : Summation in Computing

To find the total weight of all 20 students, we write $\sum_{k=1}^{20} wt[k]$.

In programming, we just add all the terms in the array together.

Using the recursive nature of the summation, we start by initializing the sum (to either 0 or $wt[1]$), then we build upon it by adding the weight of each student term by term.

The following algorithms yield the same result :

<pre>1. sum := 0 For k := 1 to 20 sum := sum + wt[k] next k</pre>	<pre>2. Sum := wt[1] For p := 2 to 20 sum := sum + wt[p] next p</pre>
--	--

↑
Initialize sum to be = 0

Then find $\sum_{k=1}^{20} wt[k]$

↑
Initialize sum to be = $wt[1]$

Then find $wt[1] + \sum_{p=2}^{20} wt[p]$

Application : Decimal to Binary Conversion

Humans use decimal notation (base 10) for convenience as we happen to have 10 fingers. On the other hand, computers use binary notation (base 2) as modern electronics use two states (0 and 1) for their signals.

For example, the number 22 is represented as follows :

Decimal Notation : $22 = 2 \times 10^1 + 2 * 10^0$

Place Value	10^4	10^3	10^2	10^1	10^0
Decimal Digits	0	0	0	2	2

So we write 22_{10} .

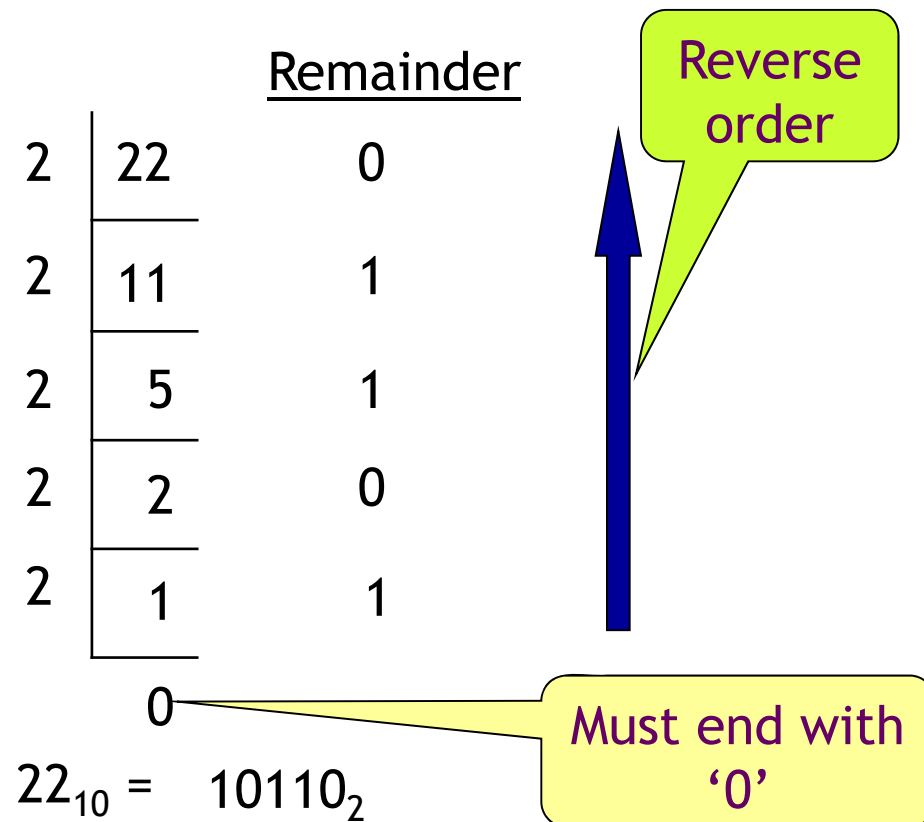
Binary Notation : $22 = 1 \times 2^4 + 0 * 2^3 + 1 * 2^2 + 1 * 2^1 + 0 * 2^0$

Place Value	2^4	2^3	2^2	2^1	2^0
Decimal Digits	1	0	1	1	0

So we write 10110_2 .

Application : Decimal to Binary Conversion (Cont'd)

The procedure for converting a decimal number (e.g. 22_{10}) to a binary number is as follows :



Application : Decimal to Binary Conversion (Cont'd)

Taking the remainder at each step as a term in a sequence, we can represent them as an array : $r[1]$, $r[2]$, $r[3]$, $r[4]$, $r[5]$.

		<u>Remainder</u>	
2	22	0	$r[1]$
2	11	1	$r[2]$
2	5	1	$r[3]$
2	2	0	$r[4]$
2	1	1	$r[5]$
	0		End Loop

$$22_{10} = 10110_2$$

$r[5] \ r[4] \ r[3] \ r[2] \ r[1]$

Application : Decimal to Binary Conversion (Cont'd)

The algorithm for this conversion, which applies sequence (arrays), functions (quotient and mod) and logic (used in WHILE loop), is as follows :

		<u>Remainder</u>	
2	22	0	r[1]
2	11	1	r[2]
2	5	1	r[3]
2	2	0	r[4]
2	1	1	r[5]
	0		End Loop

$22_{10} = 10110_2$
 r[5] r[4] r[3] r[2] r[1]

Input : DecNum (integer ≥ 0)

Algorithm :

quotient := DecNum , k := 1

While quotient $\neq 0$ or k=1

 r[k] := quotient **mod** 2

 quotient := quotient **div** 2

 k:= k+1

End while

Output :

DecNum = r[k-1]r[k-2] ... r[3]r[2]r[1]

Quotient=22, k=1

Remainder of
 $22 \div 2 = 0 = r[1]$

Set quotient as 11

Increment index by 1

Repeat loop till
quotient =0

Print in reverse

Part 4 Recursive Sequence

Section 1 Introduction

- Introduction
- Components of Recursive Formula

Introduction

In some situations, it may not always be possible to find an explicit formula to represent a sequence directly.

An alternative is to define the sequence using recursion.

Recursion is a method in computer science where a solution to a problem depends on the solution to a smaller version of the same problem.

A recursive sequence would usually have a few initial terms that are given, and later terms of the sequence are defined by relating them to a few of the earlier terms.

Components in Recursive Formula

A recursive formula “must” consists of 2 components :

- (1) An equation called **recurrence relation** is required to relate later terms in the sequence to earlier terms.
- (2) It comes with **initial conditions** which are the values of the first few terms in a sequence.

For example define a sequence $b_0, b_1, b_2, \dots$ recursively as follows:

For integers $k \geq 2$, $b_k = b_{k-1} + b_{k-2}$ *recurrence relation*

$b_0 = 2, b_1 = 4$ *initial conditions*

Part 4 Recursive Sequence

Section 2

Finding Terms in Recursive Sequence

- Examples
- Exercises

Example (Finding Terms of Recursive Sequence)

Define a sequence $b_0, b_1, b_2, \dots$ recursively as follows:

$$\begin{array}{ll} \text{For integers } k \geq 2, & b_k = b_{k-1} + b_{k-2} \quad \text{recurrence relation} \\ & b_0 = 2, \quad b_1 = 4 \quad \text{initial conditions} \end{array}$$

Find the next 4 terms of the sequence.

Solution:

$$k = 2, \quad b_2 = b_1 + b_0 = 4 + 2 = 6$$

$$k = 3, \quad b_3 = b_2 + b_1 = 6 + 4 = 10$$

$$k = 4, \quad b_4 = b_3 + b_2 = 10 + 6 = 16$$

$$k = 5, \quad b_5 = b_4 + b_3 = 16 + 10 = 26$$

Exercise 1 (Finding Terms of Recursive Sequence)

Define a sequence $a_0, a_1, a_2, \dots$ recursively as follows:

For integers $n \geq 2$, $a_n = a_{n-1} + n a_{n-2}$ *recurrence relation*
 $a_0 = 1, a_1 = 2$ *initial conditions*

Find a_2, a_3 and a_4 .

Solution:

$$n = 2, \quad a_2 = a_1 + 2a_0 = 2 + 2(1) = 4$$

$$n = 3, \quad a_3 = a_2 + 3a_1 = 4 + 3(2) = 10$$

$$n = 4, \quad a_4 = a_3 + 4a_2 = 10 + 4(4) = 26$$

Example (Change of Initial Conditions)

Let $a_1, a_2, a_3, \dots$ and $b_1, b_2, b_3, \dots$ satisfy the recurrence relation as shown below for all integers $n \geq 2$.

$$\begin{aligned}a_n &= 4a_{n-1} \\ a_1 &= 1\end{aligned}$$

Find a_2, a_3, a_4 .

Solution:

$$n = 2, \quad a_2 = 4a_1 = 4$$

$$n = 3, \quad a_3 = 4a_2 = 16$$

$$n = 4, \quad a_4 = 4a_3 = 64$$

$$\begin{aligned}b_n &= 4b_{n-1} \\ b_1 &= 2\end{aligned}$$

Find b_2, b_3, b_4 .

Solution:

$$n = 2, \quad b_2 = 4b_1 = 8$$

$$n = 3, \quad b_3 = 4b_2 = 32$$

$$n = 4, \quad b_4 = 4b_3 = 128$$

The sequences are different (due to changes in initial conditions).

Exercise 2 (Finding Terms of Recursive Sequence)

Find the next 3 terms of the recursively defined sequence

$$T_1 = 5, \quad T_n = T_{n-1} + 3$$

Solution:

$$T_1 = 5 \quad (\text{given})$$

$$T_2 = T_1 + 3 = 5 + 3 = 8$$

$$T_3 = T_2 + 3 = 8 + 3 = 11$$

$$T_4 = T_3 + 3 = 11 + 3 = 14$$

The next 3 terms are 8, 11, 14.

Exercise 3 (Finding Terms of Recursive Sequence)

Find the first 5 terms of the recursively defined sequence

$$T_1 = 2, \quad T_2 = 5, \quad T_n = 2T_{n-1} - T_{n-2}$$

Solution:

$$T_1 = 2 \quad (\text{given})$$

$$T_2 = 5 \quad (\text{given})$$

$$T_3 = 2T_2 - T_1 = 2(5) - 2 = 8$$

$$T_4 = 2T_3 - T_2 = 2(8) - 5 = 11$$

$$T_5 = 2T_4 - T_3 = 2(11) - 8 = 14$$

The first 5 terms are 2, 5, 8, 11 and 14.

Part 4 Recursive Sequence

Section 3

Finding Recursive Formula

- Why The Need?
- The Rabbit Problem and Fibonacci Sequence
- Examples and Exercises

Why The Need?

In some situations, the event occurs in a recurring manner, and builds upon earlier event(s).

The best way to capture such situations is via the use a recursive formula.

A recursive sequence would usually have a few initial terms that are given, and later terms of the sequence are defined by relating them to a few of the earlier terms.

Introduction to The Fibonacci Sequence

In AD1225, **Fibonacci** (more correctly known as Leonardo da Pisa) participated in a mathematics tournament at Pisa (ordered by the emperor Federick II) to solve a problem about the reproduction of rabbits.

It was while deriving at the solution to this problem that he found the recursive sequence that was later named after him.

Rabbit Problem

Let's begin with 1st month ...

1 single pair of rabbit



No. of pairs

1

How many pairs of rabbit will there be in a year?

Assuming that ...

1. Each pair produces a **new pair** each month
2. The **new pair** becomes productive from the **2nd** month onwards
3. No rabbits die in the process.

Rabbit Problem (Cont'd)

Then in 2nd month ...

Still 1 pair only

(since each **new pair** becomes
productive from the **2nd**
month onwards)



No. of pairs

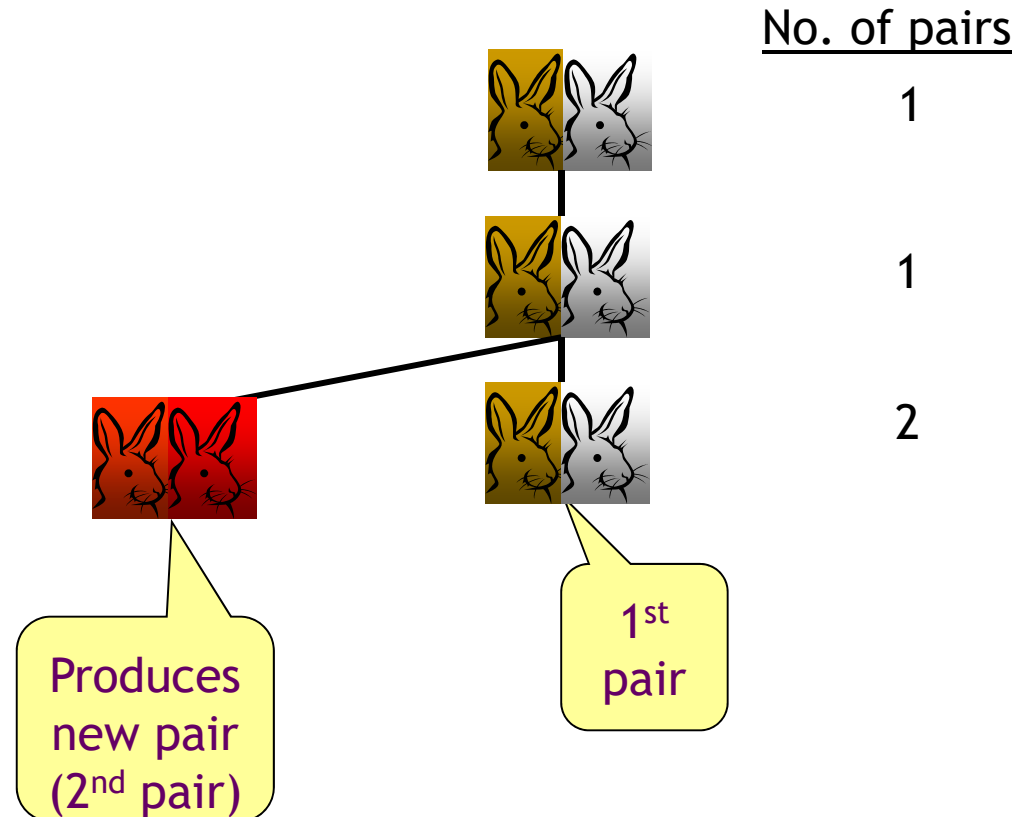
1

1

Rabbit Problem (Cont'd)

Then in 3rd month ...

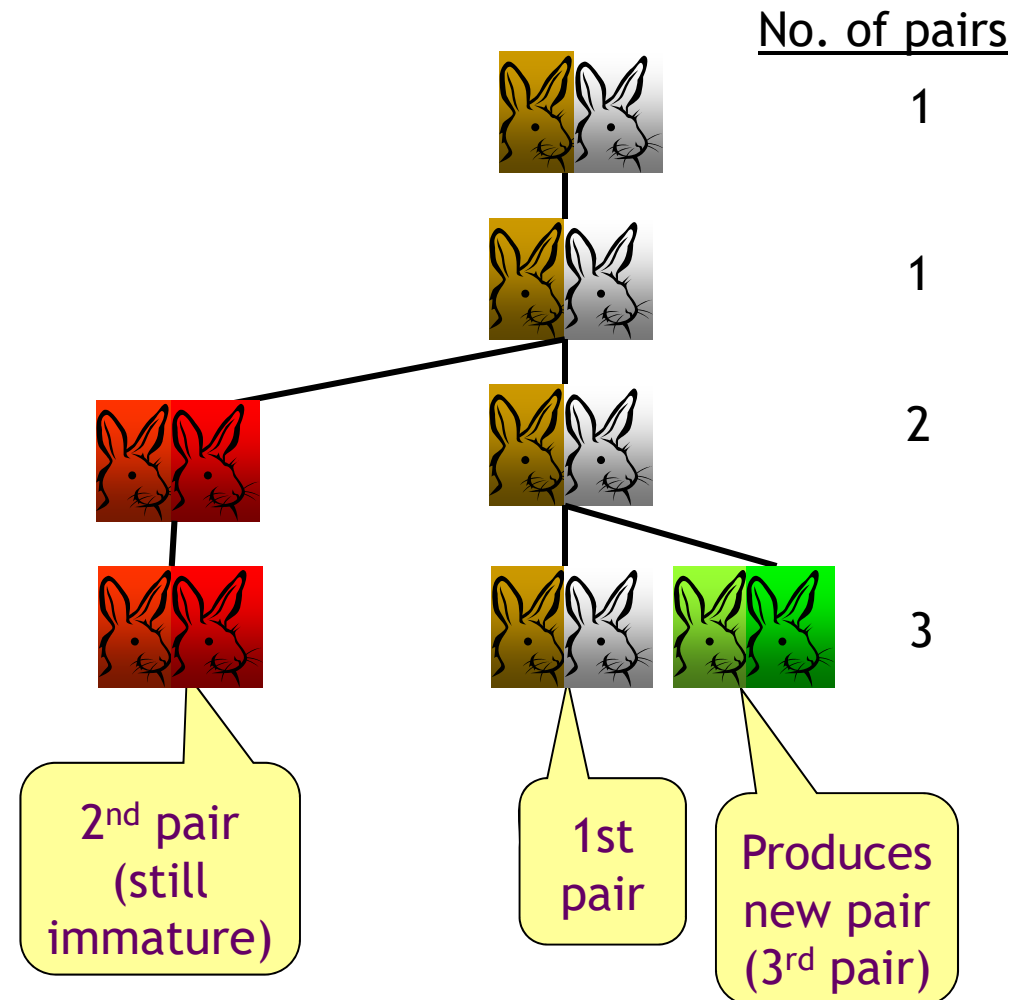
Becomes 2 pairs



Rabbit Problem (Cont'd)

Then in 4th month ...

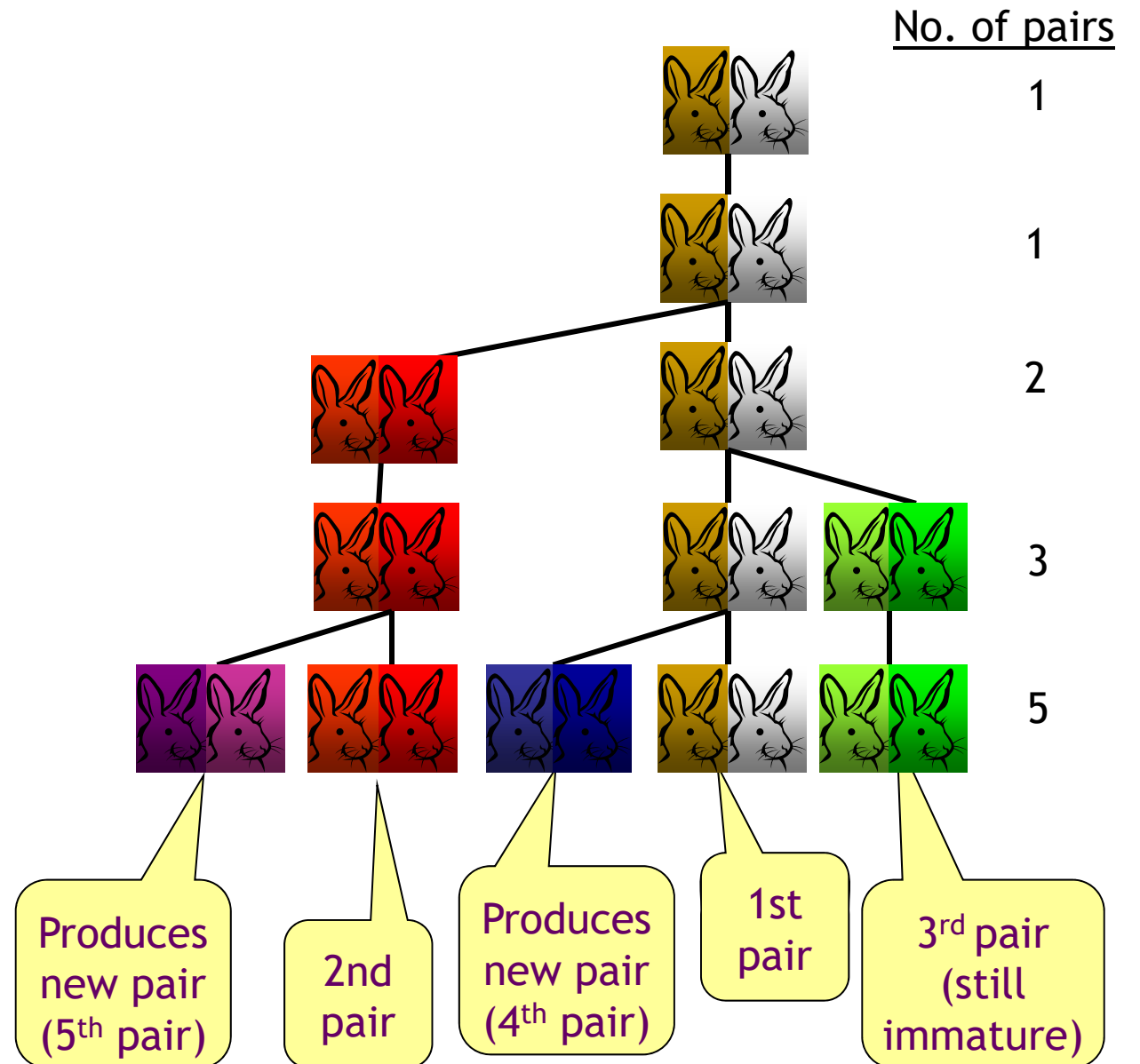
Becomes 3 pairs



Rabbit Problem (Cont'd)

Then in 5th month ...

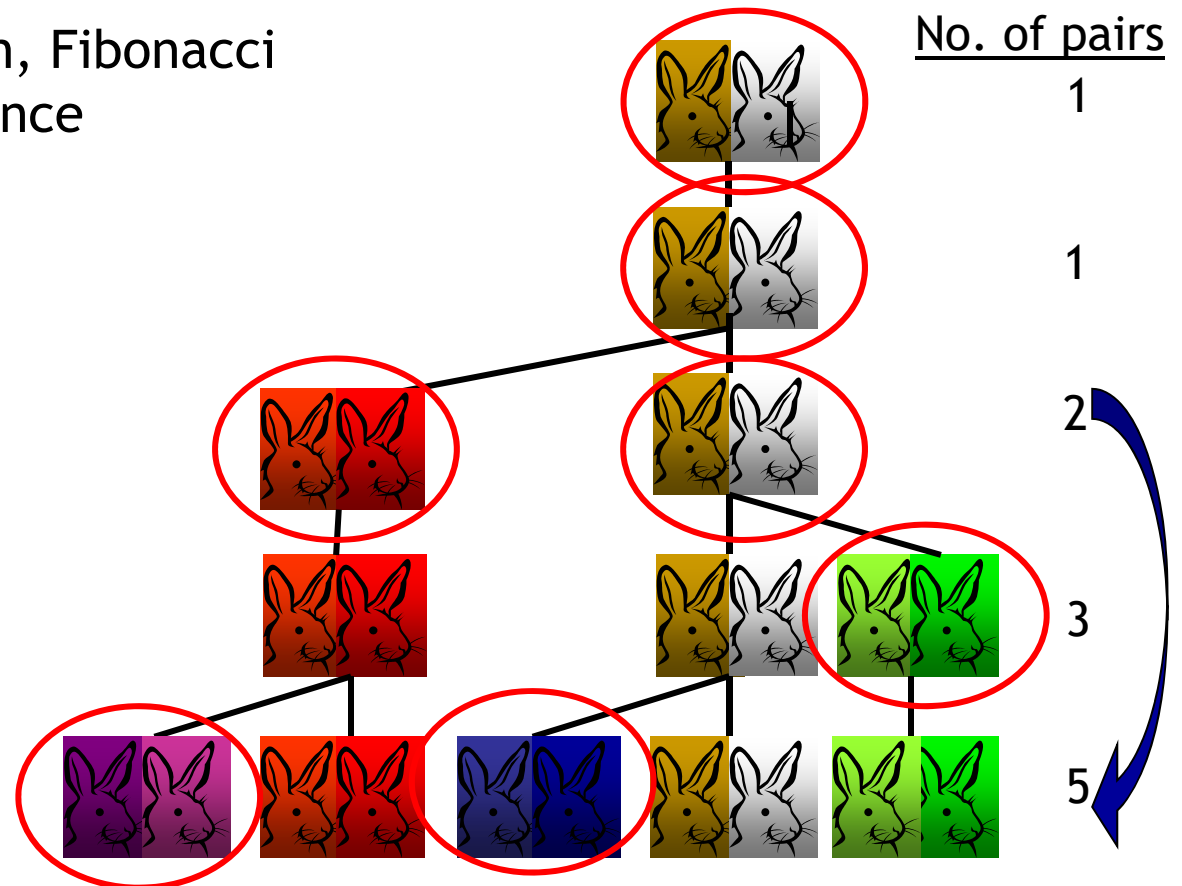
Becomes 5 pairs



Rabbit Problem (Cont'd)

While deriving the solution, Fibonacci found this recursive sequence

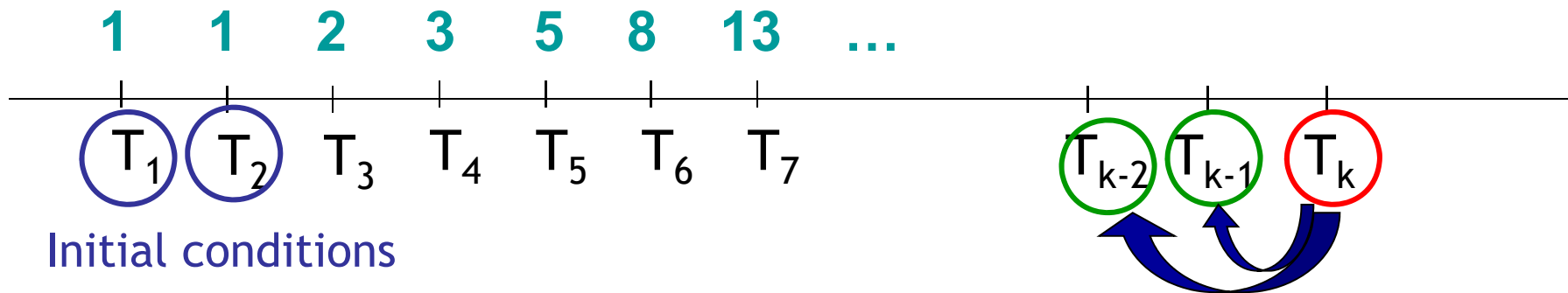
1 1 2 3 5 8 13 ...



No. of rabbit pairs **alive** at the end of month **k** = No. of rabbit pairs **alive** at the end of month **k-1** + No. of rabbit pairs **born** at the end of month **k**

No. of rabbit pairs **alive** at the end of month **k-1** = No. of rabbit pairs **alive** at the end of month **k-1** + No. of rabbit pairs **alive** at the end of month **k-2**

Fibonacci Sequence



No. of rabbit pairs alive at the end of month k	=	No. of rabbit pairs alive at the end of month $k-1$	+	No. of rabbit pairs alive at the end of month $k-2$
T_k	=	T_{k-1}	+	T_{k-2}

$$T_1 = 1$$

Initial condition

$$T_2 = 1$$

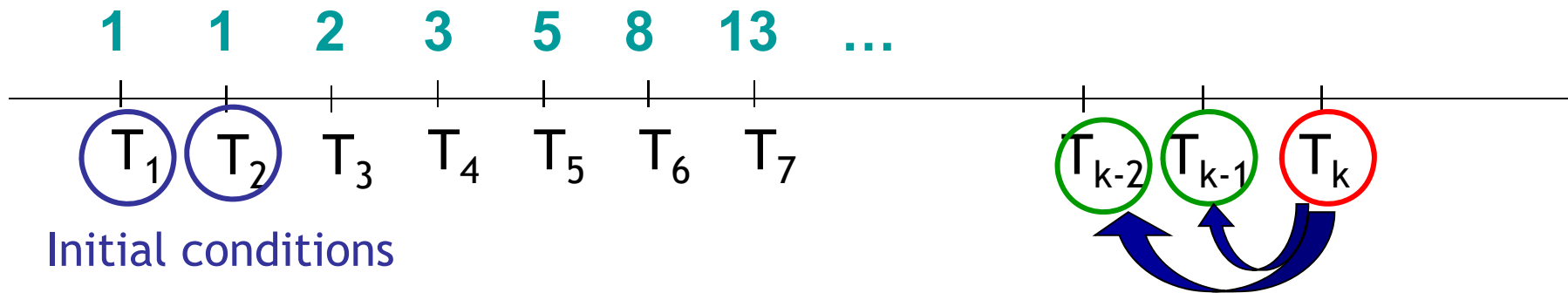
Initial condition

$$T_k = T_{k-1} + T_{k-2}$$

Recurrence Relation

**Recursive
Formula**

Find The Next 2 Terms in Fibonacci Sequence



$$T_1 = 1$$

Initial condition

$$T_2 = 1$$

Initial condition

$$T_k = T_{k-1} + T_{k-2}$$

Recurrence Relation

**Recursive
Formula**

$$T_8 = T_{8-1} + T_{8-2} = T_7 + T_6 = 13 + 8 = 21$$

$$T_9 = T_{9-1} + T_{9-2} = T_8 + T_7 = 21 + 13 = 34$$

Example (Finding Recursive Formula)

A newly married couple intends to buy an apartment through a walk-in promotion. They are looking through the price list of a particular eight-storey block.

An apartment on the 1st floor costs \$475 000.

For every increase in floor level, the price increases by \$12 000.

Write the recursive formula.

How much will an apt on the 4th floor cost?

Solution:

$$T_1 = 475\,000 \qquad T_n = T_{n-1} + 12\,000$$

$$T_2 = T_1 + 12\,000 = 475\,000 + 12\,000 = 487\,000$$

$$T_3 = T_2 + 12\,000 = 487\,000 + 12\,000 = 499\,000$$

$$T_4 = T_3 + 12\,000 = 499\,000 + 12\,000 = 511\,000$$

The cost of the fourth floor is \$511 000.

Exercise (Finding Recursive Formula)

An IT firm offered you a job on 1/1/2004 with starting pay of \$18000 / yr. The firm promised to give a guaranteed annual increment of \$950.

- (i) Write a recursive formula to compute your annual salary based on what the IT firm has promised you.
- (ii) Find your annual salary for the first 3 years.

Solution:

$$(i) \quad T_1 = 18000 \quad T_n = T_{n-1} + 950$$

$$(ii) \quad T_1 = 18\,000$$

$$T_2 = T_1 + 950 = 18000 + 950 = 18950$$

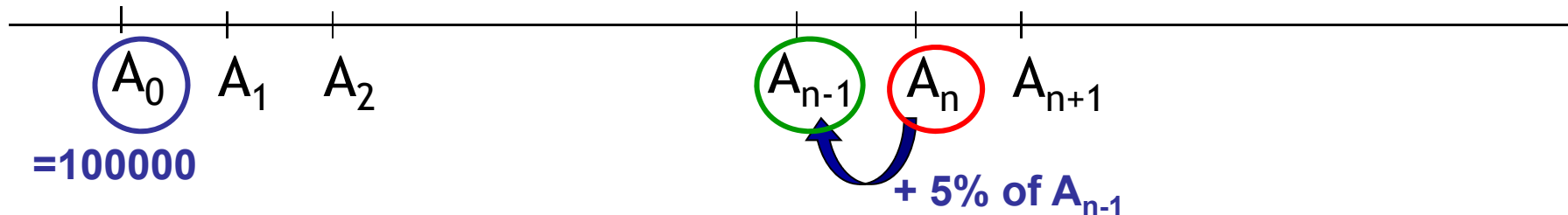
$$T_3 = T_2 + 950 = 18950 + 950 = 19900$$

The annual salary for the first 3 years : \$18000, \$18950, \$19900

Application : Compound Interest

- On your 21st birthday, you receive a letter informing you that :
- On your 18th birthday, your grandpa had deposited \$100,000 in a bank,
- earning 5% interest compounded annually.
- He intends to transfer the ownership of this account to you provided you can calculate the amount currently in the account.

Amount in account at the end of the year	=	Amount in account at the end of previous year	+	(0.05)	Amount in account at the end of previous year
---	---	--	---	--------	--



18th : $A_0 = 100\ 000$

19th : $A_1 = (1.05)(100\ 000) = 105000$

20th : $A_2 = (1.05)(105\ 000) = 110250$

21st : $A_3 = (1.05)(110\ 250) = 115762.5$

$A_n = A_{n-1} + (0.05)A_{n-1}$ $A_n = 1.05A_{n-1}$

Application : Divide and Conquer Algorithm

- Divide and Conquer algorithm works as follows :
 - Divide a complex problem into smaller instances of itself;
 - Conquer the smaller (but easier) problems recursively;
 - Combine the answers to these smaller problems so as to solve the original, more complex problem.
- For example :
 - Sort a list of student names into alphabetical order
 - Sort the students by their exam score
- To do that,
 - Divide the list into 2 smaller lists;
 - Conquer each of the 2 smaller lists recursively;
 - Combine the sorted smaller lists back together recursively.

Application : Merge Sort Algorithm

- Merge Sort is an efficient sorting algorithm that uses the divide and conquer algorithm to sort an array.

- To sort the following numbers by merge sort :

8	3	6	5
---	---	---	---

- Divide :

8	3
---	---

6	5
---	---

- Sub-divide further :

8	3
---	---

6	5
---	---

- Sort 1st sub-list :

3	8
---	---

 Then sort 2nd sub-list :

5	6
---	---

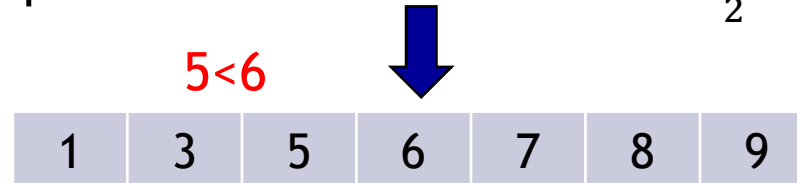
- Merge the 2 sub-lists :

3	5	6	8
---	---	---	---

Application : Binary Search Algorithm

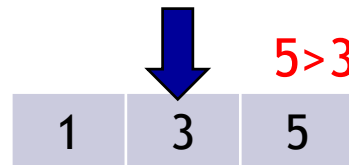
- Binary Search is an efficient search algorithm that uses the divide and conquer algorithm to search for an item in a sorted array.
- To search for '5' in this sorted array :

- Find middle position = floor of $\frac{\text{highest index} + \text{lowest index}}{2} = \lfloor \frac{7+1}{2} \rfloor = 4^{\text{th}}$



- Compare : Since $5 < 6$, take only the left sub-list :

- Find middle position = floor of $\frac{\text{highest index} + \text{lowest index}}{2} = \lfloor \frac{3+1}{2} \rfloor = 2^{\text{nd}}$



- Compare : Since $5 > 3$, take only the right sub-list :

5

Part 5 Solving Recurrence Relations by Iteration

Section 1 Introduction

- Why The Need?
- Method of Iteration

Why The Need?

From Recursive Formula To Explicit Formula

Suppose you have a sequence that satisfies a certain recurrence relation and initial conditions.

When you need to compute terms with very large subscripts (e.g. 100th term), you will need to run through 100 iterations, which is very tedious.

It is **more efficient** find an **explicit formula**, in place of the recursive formula, so as to calculate the terms **directly**.

Such an explicit formula is called a **solution** to the recurrence relation.

The most basic method to find explicit formula for a recursive sequence is the **method of iteration**.

Method of iteration

Given the sequence $a_0, a_1, a_2, \dots$ defined by a recurrence relation and initial conditions.

Step 1: Start from initial conditions

Step 2: Calculate successive terms until you see a pattern developing

Step 3: Guess an explicit formula

Part 5 Solving Recurrence Relations by Iteration

Section 2 Method of Iteration

- Method of Iteration
- Solve for Explicit Formula

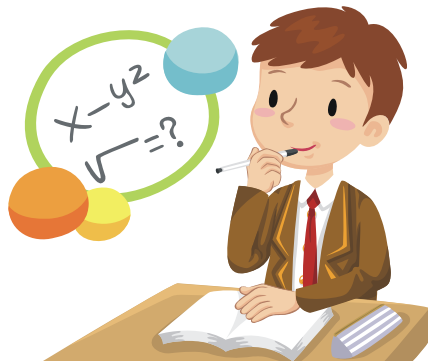
Method of iteration

Given the sequence $a_0, a_1, a_2, \dots$ defined by a recurrence relation and initial conditions.

Step 1: Start from initial conditions

Step 2: Calculate successive terms until you see a pattern developing

Step 3: Guess an explicit formula



Example (Solve for Explicit Formula)

Let $a_0, a_1, a_2, \dots$ be the sequence defined recursively as follows:

$$\begin{array}{ll} \text{For integers } n \geq 1, & a_n = a_{n-1} + 3 \quad \text{recurrence relation} \\ & a_0 = 1 \quad \text{initial condition} \end{array}$$

Use iteration to guess an explicit formula for the sequence.

Step 1 : Start from initial conditions

$$a_0 = 1 = 1 + (0)(3)$$

$$\text{Step 2 : } a_1 = a_0 + 3 = 1 + 3 = 1 + (1)(3)$$

$$a_2 = a_1 + 3 = (1 + 3) + 3 = 1 + (2)(3)$$

$$a_3 = a_2 + 3 = (1 + 3 + 3) + 3 = 1 + (3)(3)$$

$$a_4 = a_3 + 3 = (1 + 3 + 3 + 3) + 3 = 1 + (4)(3)$$

Step 3 : Guess the explicit formula

$$a_n = 1 + 3n$$

Exercise (Solve for Explicit Formula)

Let $a_0, a_1, a_2, \dots$ be the sequence defined recursively as follows:

$$\begin{aligned} \text{For integers } n \geq 1, \quad a_n &= r a_{n-1} && \text{recurrence relation} \\ a_0 &= a && \text{initial condition} \end{aligned}$$

Use **iteration** to guess an explicit formula for the sequence.

Step 1 : Start from initial conditions

$$a_0 = a = r^0 a$$

$$\text{Step 2 : } a_1 = r a_0 = r a = r^1 a$$

$$a_2 = r a_1 = r(r a) = r^2 a$$

$$a_3 = r a_2 = r(r^2 a) = r^3 a$$

$$a_4 = r a_3 = r(r^3 a) = r^4 a$$



Step 3 : Guess the explicit formula

$$a_n = r^n a$$

Part 5 Solving Recurrence Relations by Iteration

Section 3 Special Recurrence Relation

- Special Recurrence Formulas
- Quick Explanation of Formulas
- Solve Explicit Formula and Simplify

Special Recurrence Formulas

Explicit formulas obtained by iteration can sometimes be **simplified** using the following special recurrence formulas.

$$1 + r + r^2 + r^3 + \dots + r^n = \frac{r^{n+1} - 1}{r - 1} \text{ for all integers } n \geq 0.$$

$$1 + 2 + 3 + \dots + n = \frac{n(n + 1)}{2} \text{ for all integers } n \geq 1.$$

Quick Explanation of the Special Recurrence Formula 1

Quick explanation as to why

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2} \quad \text{for all integers } n \geq 1.$$

Suppose you want to find the sum of all numbers from 1 to 100.

$$\begin{array}{cccccccccccccccc} 1 & + & 2 & + & 3 & + & 4 & + & \dots & + & 97 & + & 98 & + & 99 & + & 100 \\ 100 & + & 99 & + & 98 & + & 97 & + & \dots & + & 4 & + & 3 & + & 2 & + & 1 \\ 101 & & 101 & & 101 & & 101 & & & & 101 & & 101 & & 101 & & 101 \end{array}$$

$$1 + 2 + 3 + \dots + 100 = \frac{100(101)}{2} = \frac{10100}{2} = 5050$$

Quick Explanation of the Special Recurrence Formula 2

Quick explanation as to why

$$1 + r + r^2 + r^3 + \dots + r^n = \frac{r^{n+1} - 1}{r - 1} \quad \text{for all integers } n \geq 0.$$

$$\text{Let } S_n = 1 + r + r^2 + r^3 + \dots + r^n \quad \text{-----} \textcircled{1}$$

Multiply S_n by r

$$\begin{aligned} rS_n &= r(1 + r + r^2 + r^3 + \dots + r^n) \\ &= r + r^2 + r^3 + \dots + r^n + r^{n+1} \quad \text{-----} \textcircled{2} \end{aligned}$$

$$rS_n - S_n \quad \text{-----} \textcircled{2} - \textcircled{1}$$

$$= (r + r^2 + \dots + r^n + r^{n+1}) - (1 + r + r^2 + \dots + r^n)$$

$$= r^{n+1} - 1 \quad \text{-----} \textcircled{3}$$

Quick Explanation of the Special Recurrence Formula 2

Also, $rS_n - S_n = (r - 1)S_n$ ----- take out common factor of S_n

and $rS_n - S_n = r^{n+1} - 1$ ----- from ③

Since LHS is the same, equate RHS to get :

$$(r - 1)S_n = r^{n+1} - 1$$

$$S_n = \frac{r^{n+1} - 1}{r - 1} \text{ ----- divide both sides by } (r - 1)$$

$$1 + r + r^2 + r^3 + \dots + r^n = \frac{r^{n+1} - 1}{r - 1} \text{ ----- using ①}$$

Example

(Solve for Explicit Formula & Simplify)

$$1 + r + r^2 + r^3 + \dots + r^n = \frac{r^{n+1} - 1}{r - 1}$$

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

Let $a_1, a_2, \dots$ be the sequence defined recursively as follows:

For integers $n \geq 1$, $a_n = a_{n-1} + 2n$ *recurrence relation*
 $a_1 = 2$ *initial condition*

Use iteration to guess an explicit formula and simplify it.

Step 1:
Start with initial conditions $a_1 = 2$

Step 2:

Calculate successive terms
until you see a pattern
developing

$$a_2 = a_1 + 2(2) = 2(1) + 2(2) = 6$$

$$a_3 = a_2 + 2(3) = 2(1) + 2(2) + 2(3) = 12$$

$$a_4 = a_3 + 2(4) = 2(1) + 2(2) + 2(3) + 2(4) = 20$$

$$a_5 = a_4 + 2(5) = 2(1) + 2(2) + 2(3) + 2(4) + 2(5) = 30$$

Step 3:

Guess explicit formula

$$a_n = 2(1) + 2(2) + 2(3) + \dots + 2(n)$$

$$= 2(1 + 2 + 3 + \dots + n) = 2 \left(\frac{n(n+1)}{2} \right) = n(n+1)$$

simplify

Exercise

(Solve for Explicit Formula & Simplify Answer)

Let $a_0, a_1, a_2, \dots$ be the sequence defined recursively as follows:

$$\begin{array}{ll} \text{For integers } n \geq 1, & a_n = a_{n-1} + 10 \quad \text{recurrence relation} \\ & a_0 = 200 \quad \text{initial condition} \end{array}$$

Use iteration to guess an explicit formula and simplify it.

Step 1:
Start with initial conditions $a_0 = 200$

Step 2:
Calculate successive terms until you see a pattern developing

$$\begin{array}{llll} a_1 = a_0 + 10 = 200 + 10 & = 210 \\ a_2 = a_1 + 10 = 200 + 10 + 10 & = 220 \\ a_3 = a_2 + 10 = 200 + 10 + 10 + 10 & = 230 \\ a_4 = a_3 + 10 = 200 + 10 + 10 + 10 + 10 & = 240 \end{array}$$

Step 3:
Guess explicit formula

$$a_n = 200 + \underbrace{10 + 10 + \dots + 10}_{n \text{ times}} = 200 + 10n$$

simplify

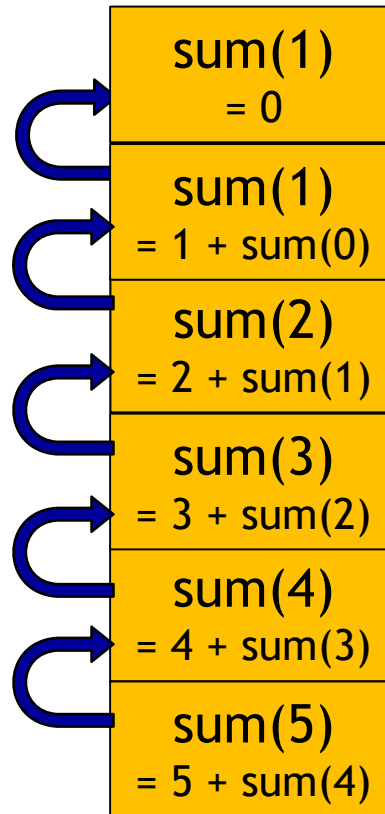
Application : Recursion vs Iteration in Coding

These two algorithms can both be used to sum the integers 1, 2, 3, ..., n.
What is the difference between them?

Recursion

```
function sum(n) {  
  If n <= 0  
    return 0  
  else  
    return n + sum(n-1)  
}
```

Function calls a
smaller version of
itself repeatedly



Iteration

```
function sum(n) {  
  sum := 0  
  For k := 1 to n  
    sum := sum + k  
  next k  
  return sum  
}
```

Use loop to run
statement
repeatedly

Sum
= sum + k

Application : Recursion vs Iteration

Similarities

	Recursion	Iteration
Basic	Executes a set of statements repeatedly	Executes a set of statements repeatedly

Differences

	Recursion	Iteration
Basic	A function calls the function itself repeatedly until a certain base condition is reached	A set of instructions is repeated executed using a loop until a certain condition is reached
Execution Speed	Slower	Faster
Length of Code	Shorter, Easier to read	Longer

*****END OF TOPIC*****